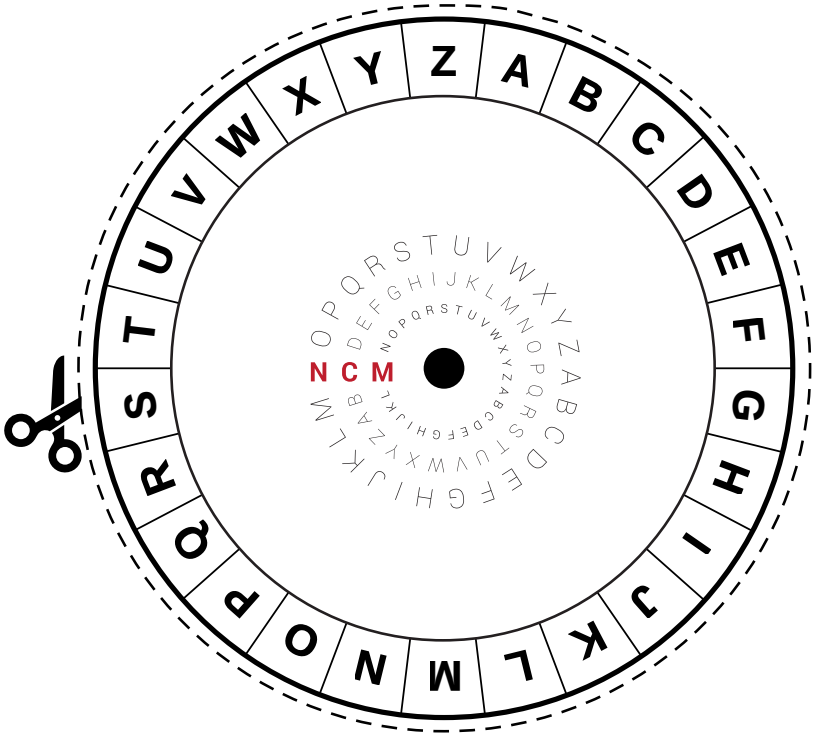
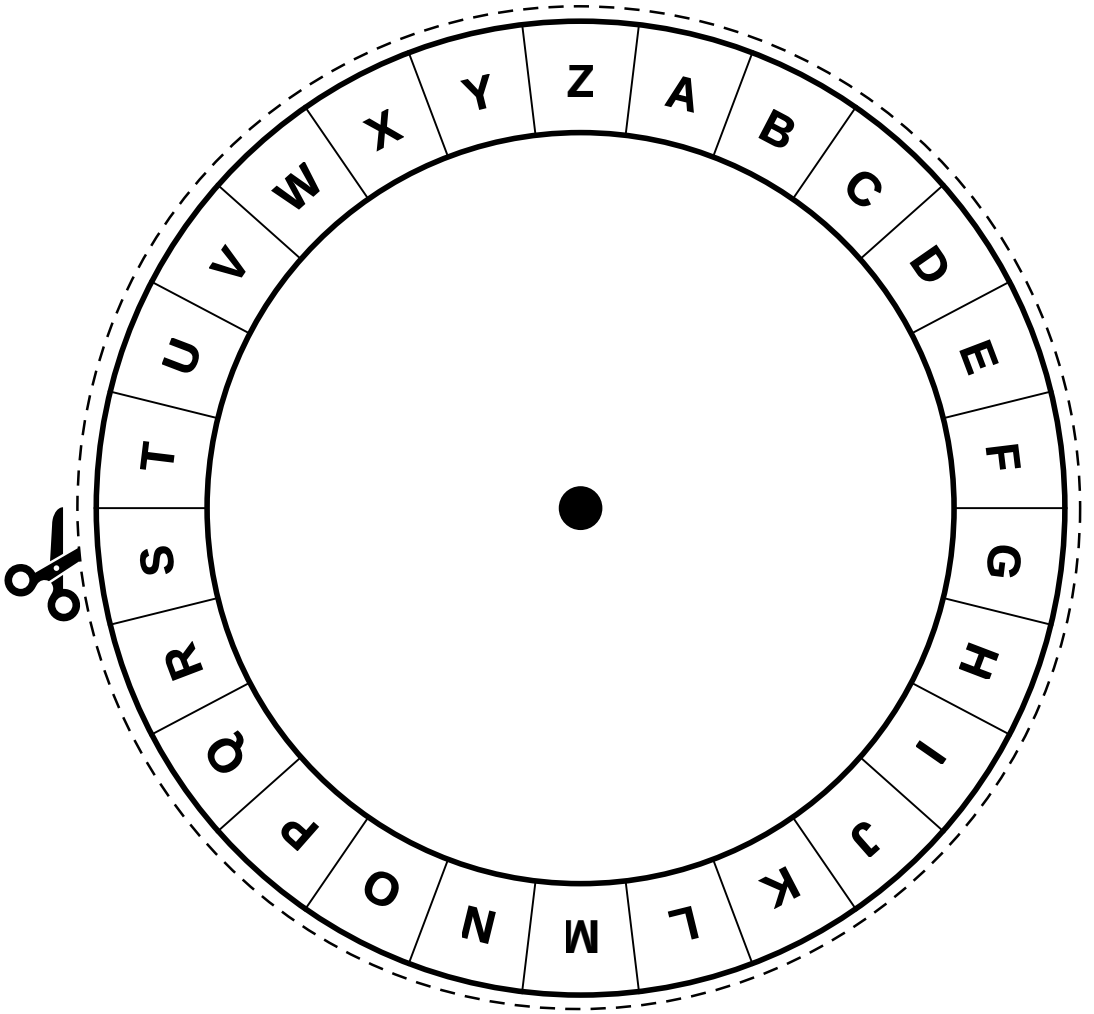
MAKING A CIPHER WHEEL: **TEMPLATES**

INSTRUCTIONS CONTINUE ON NEXT PAGE





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MAKING A CIPHER WHEEL: **HOW TO MAKE**

YOU WILL NEED

- This template
- Scissors
- Brass fastener (or tack and small eraser)
- A friend to make secret messages with (send them this template!)

HOW TO BUILD

- Cut out the two circles.
- Carefully poke a hole through the dot in the center of each wheel.
- Place the smaller circle over the larger circle so that the holes line up, and join them together using a brass fastener or push a tack through the center into an eraser.
- The two circles should be able to rotate separately.

ENCRYPTING A MESSAGE

- Pick a letter from the smaller disk on top - this will be used as the KEY that will be shared between you and your friend to encrypt and decrypt the message. Once you have picked a letter from the smaller disk, line it up with the 'A' from the larger disk.

  Example Key:
  LARGE DISK: A SMALL DISK: M KEY: A=M

- Begin encrypting each letter of your secret message. Find the first letter of your message on the larger disk and write down the cipher letter which appears underneath it on the smaller disk.

  Example: A=M, B=L, C=K, etc.
  LARGE DISK: ABCDEFGHIJKLMNOPQRSTUVWXYZ
  SMALL DISK: MLKJIHGFEDCBAZYXWVUTSRQPON

- Continue finding each letter until your message has been encrypted. Once you are finished, send your friend the encrypted message and KEY separately.

  PLAIN TEXT: MAKING CODES IS FUN
  CIPHER TEXT: AMCEZG KYJIU EU HSZ

NOTE: Make sure your friend knows the KEY so they can use their Cipher Disk to break the cipher!

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